

One-child Policy, Pension Reforms and Household Saving Rates in China

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Abstract

This paper examines the impact of the one-child policy and pension reforms on household savings in China using an overlapping generations model. The model incorporates intergenerational transfers, endogenous fertility, human capital investment, and the pension system. I find that the one-child policy increases individual household saving rates across all age groups. And pension benefits discourage household savings. However, the response varies across households with different initial wealth. Besides, I find that the one-child policy increases investment in children's human capital from parents, while pension benefits have the opposite effect. However, on the aggregate level, the predictions are consistent with the micro-level pattern.

1 Introduction

In many countries, like China, people have more kids as a way to ensure old-age support. But having kids are costly and a higher number of kids is translated into less consumption and education per kid. The family plan policies and pension reforms will affect the old-age insurance. Corresponding to those changes, households will change their saving decisions and human capital investment in children for old-age support. This paper tries to identify to what extent the family plan policies and pension reforms will affect household savings and human capital accumulation in the case of China. China has introduced the One-child policy in 1979 to curb high population growth. And it is relaxed to a two-child policy in 2011. This offers a natural experiment to examine the effects. As a consequence of the One-child policy, fertility declines drastically in a short period, from three children per household in the 1960s on average to one child in the 1980s. The decrease in the number of children per family will affect the amount of old-age transfer. That means households have to save more on their own. In the meantime, the long-lasting low fertility and increasing life expectancy lead to fast population aging in China. As the traditional social norm of family support to the old is threatening by the

One-child policy, the Chinese government has taken reforms on its social security system to help secure the old-age life. A well-functioned pension system will reduce the personal incentive of savings for the old-age. From the 1960s to 2010s, China's gross saving rate has doubled and then it starts decreasing. And the average education level of Chinese has been increasing since 1960 as well. How much of the increases can be explained by those policy changes?

In this paper, I examine the effects of the One-child policy and pension reforms on household saving rates and human capital accumulation in China by extending the overlapping generations model in Choukhmane et al. (2013) with the pension system. Using this model, I investigate the micro-level impacts on individual households together with the impacts on the aggregate economy. The model incorporates the social security system, endogenous fertility, and human capital investment. While the government has taken efforts to improve the role of the public pension system, family support is still the main source of old-age support Choukhmane et al. (2013). Parents consider children as insurance of old-age life. The restriction from the One-child policy makes parents save more or invest more in one child. By jointly modeling the fertility and human capital investment choices from parents on children, I can explore the extent to which the One-child policy can increase household saving, human capital accumulation, and its potential influence of the relaxation of the One-child policy. The model also includes the social security system. The pension benefits can replace the role of children in supporting the old. So the pension reforms in China may also affect fertility decisions and human capital investment of households. On the other hand, fertility and demographic structure will affect pension benefits as well. By adding the pension system in the model, I can account for the interactions between pensions, fertility, and human capital and examine the possible influences of pension reforms in China.

A key feature of the model is the intergenerational transfer between parents and children. The parents will raise and invest in children until they become independent. On the other hand, children will give transfers to old parents once they retired. The children serve as old-age insurance. Investing in children's human capital provides an alternative to secure the old life (quantity-quality tradeoffs). The other features of the model include a pay-as-you-go pension system, productivity growth, and so on.

To examine the effects, I assume the economy starts at a steady state without the One-child policy and pension system. Then I shock the economy with the One-child policy by putting the fertility constraints and add the pension system. By comparing model results of the three scenarios, I find that the One-child policy will increase individual household's saving decisions across all age groups. And there exist quantity-quality tradeoffs. The One-child policy will lead to more investment in children's human capital. Besides, there are variations in the response to the One-child policy for households with different initial wealth. On the other hand, pension benefits will discourage personal savings and investment in children. ¹On the aggregate level, the model predicts a similar pattern.

¹Although the results cannot represent real effects without careful calibration to the real economy in China, the predictions are comparable to literature. For instance, using the main parameters from Choukhmane et al. (2013), the model predicts the One-child policy will increase the individual household saving rate by 15 percentage points from three children to one child. It is reasonable

However, the results are based on an assumption of an evenly-distributed population. To pin down the effects quantitatively, I need to calibrate the population structure. This will be the next step.

Although the implications of the model are consistent with the literature, the framework in this paper incorporates endogenous fertility and human capital decisions and pension reforms in a structural model. It captures the interactions among them. Using this framework, I can examine the potential effects of relaxation of the One-child policy and pension reforms on fertility and the Chinese economy. By calibrating it to the Chinese economy, it can give implications of the future savings changes and human capital accumulation in China.

The paper is an extension of Choukhmane et al. (2013) with the pension system. By introducing the pension system, I can examine its effects on savings and human capital considering its interaction with fertility. China has relaxed the One-child policy and taken pension reforms to cover the whole population in 2011. Using this framework, I can explore the effects of those policy changes on the economy as well. This paper is closely related to the studies of high saving rates in China. Curtis et al. (2015) shows the link between demographic transitions on China's household saving rate on the aggregate level. However, it does not account for the family support from children and linkages with micro-level data. Imrohoroglu et al. (2017) includes long-term care risks together with family supports in the economy to explain the high saving rates in China and provides evidence for the transfer channel in saving. Different from them, this framework focus on the effects of the One-child policy and fertility changes. It models the endogenous fertility and human capital investment decision, through which it can capture the interactions between fertility and the pension system. This paper also relates to research about the One-child policy and its economic effects in China. Zhang (2017) introduces the evolution of China's One-child policy and its effects on family outcomes from research. But most work focus on the micro-level effects on households. Seldom incorporates it in a structural model and examines the aggregate implications. Bairoliya et al. (2017) builds a rich dynamic general equilibrium model (DSGE) to examine the effect of the relaxation of the One-child policy on China's economy. The model includes family households with pension and health insurance, human capital investment, and intra-family transfers. Unlike them, this model assumes the family transfer exogenously given and model the fertility decisions, thus can account for the interaction between fertility and pensions. It can examine the relaxation effect on fertility given different pension systems. The work is also related to studies about pension reform and its effect on the Chinese economy. Bairoliya et al. (2018) studies the potential macroeconomic and welfare effects of raising rural pension benefit rates using a DSGE model. Song et al. (2015) carefully model different types of pension reforms and examine their potential welfare effects across generations in China using an overlapping generations model. Unlike those work, this paper focuses more on the effects of fertility changes and their interaction with the pension system.

One caveat about modeling the intergenerational transfer is that I assume it is exoge-

comparing with the empirical estimation of 7-10 percentage points increase in saving rate for twins' family in Choukhmane et al. (2013).

nous. There are two types of intergenerational transfer to the old in this model: private and public. I assume both of them are exogenous. This makes it easier to model fertility choice and investment in children. The private transfer is treated as a social norm and exogenously given. It works for the past decades as China's social security system is not well-functioned. However, it may affect future prediction with the improvement of the social security system and the interactions between the private and public support. Another thing to note is that I abstract from the general equilibrium effects through interest rates and wages. The interest rates facing households are relatively stable over time due to the financial control in China. This makes the partial equilibrium assumption to be realistic in this context. The theoretical analysis of general equilibrium is well-explained in Banerjee et al. (2014).

The rest of the paper is organized as follows. Section 2 introduces the background information and motivates the assumptions in the model. Section 3 builds up the model used to examine the impacts of the policies. Section 4 shows the parameterization of the model. Section 5 discusses the micro-level and aggregate implications from the model. Section 6 summarizes and talks about future work.

2 Background

In this section, I discuss the background information about the One-child policy and demographics, pension system, savings, and human capital in China.

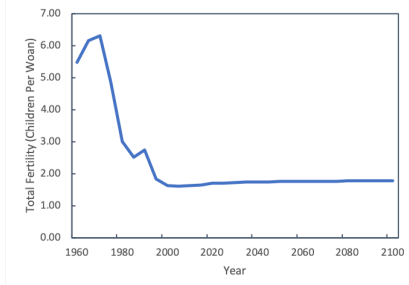
2.1 One-child Policy and China's Demographic Transitions

The concept of family planning was brought up in the 1960s facing the high population growth during the baby boom period of the 1950s-1960s. Since 1971, the Chinese government initiated the mild family planning policy, known with the slogan "later, longer, fewer" to encourage people to postpone marriage and fertility, lengthen birth spacing between children, and have fewer children. On the aggregate level, the fertility rate is affected by that policy and starts decreasing since 1970. The One-child policy is announced to be one of the basic national policies in China and was strictly enforced across the whole country in 1979. However, enforcement of the policy varies across regions in China. For example, there are exceptions for families to have a second child in the rural area. And minorities are not restricted by the policy.

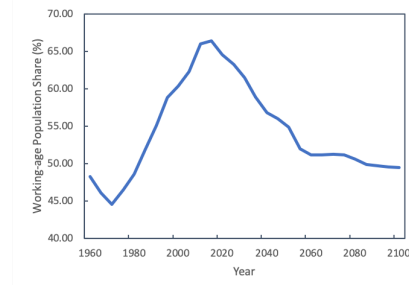
Due to those birth control policies in the 1970s, the aggregate fertility rate fell from about 6.3 per woman in 1970 to 1.6 per woman in 2000, which is below the replacement rate. There are two opposite directions in which fertility rates will affect the demographic structure. A decrease in fertility rate will reduce the number of young dependent per worker and increase the working-age population share. On the other hand, lower fertility, in the long run, will increase the number of old dependents or retirees per worker. Therefore, it will decrease the working-age population share. The net effect of fertility rates on the working-age population share is determined by the relative strengths of the two forces.

Figure 1: Demographic Transition in China: 1960-2100

(a) Fertility



(B) Working-age Population



Data Source: UN. (2019). World population prospects 2019.

Figure 1 shows the demographic transitions in China from 1960 to 2100 estimated by UN World Population Prospects 2019. The left panel shows the trend of fertility rate. Due to the one-child policy and its primitive versions, there is a sharp decline in fertility rate since 1970. It fell below the replacement rate (2.0) in the late 1990s and stays below that since then. In the beginning, the working-age population increases with this fertility decline. As shown in the right panel, the working-age population started increasing in 1970 because the number of young dependents reduces after the one-child policy. It falls from 40% of the total population in the 1970s to about 20% in 2015. After the steep rise of the working population, it is projected to decline when the one-child generation is moving into the workforce. This change in the working-age population is mainly due to the rapid increase in the number of old-age dependents. The low fertility rate and increasing old-age population share lead to the population aging problem in China.

The one-child policy is credited with cutting the population growth of China. And there is less controversy about its contribution to the rapidly growing aging population. Worried about this problem, China started to relax the one-child policy in 2013 for some families to boost the fertility rate. It allows the family formed by the one-child generations to have a second child. And in 2015, China moved to the universal two-child policy to raise the total fertility rate to 1.8 per woman by 2030 (Zeng and Hesketh, 2016). The relaxation may relieve but cannot solve the population aging problem in China. According to the UN World Population Prospects 2019, the number of old-age (65+) population keeps increasing and it accounts for more than 30% of the total population by 2100.

To summarize, the One-child policy is introduced across the whole country in 1979. Before that, the Chinese government started an informational campaign to reduce the fertility rate in 1970. And after 2015, the One-child policy is relaxed to a two-child policy.

2.2 Pension System

The current pension system in China is originally formed from 1986 and went through a major reform in 1997. Before 1986, China built up an enterprise-based pension system. Firms, mainly state-owned firms, are responsible for paying pensions to their retired employees. This fragmented pension system is financially vulnerable when China is transformed into a market-based economy, as firms can go bankrupt and workers can change jobs. Since 1986, the government decided to pool the risk of pension funds and started basing the pension on social insurance to cover all the employees. In the 1986 reform, China introduced a defined benefits system administrated by the municipalities. However, this pension system still suffered from financial distress, mainly because the enterprises can not fulfill their obligations to pay their contribution to the pension for their workers.

China took a reform of its pension system in 1997 again to tackle those issues. The management of the pooling pension funds is moved from the prefectural level to the provincial level according to the plan. It strengthened the enforcement of social security contributions from firms. And the replacement rates for future retirees are reduced as well. The 1997 pension system has two tiers, a basic pension plan, and an individual account system. The basic pension plan is transfer-based and administrated by provinces. The individual account requires a mandatory contribution from employees. However, the individual accounts are largely empty because the funds are used to supplement the cash flow deficits in the pooling account (Sin and Yu, 2005; Yuan, 2020). There are several other reforms on this pension system after 1997. The framework keeps the same. Only benefits level or contribution rates change and it accounts for more realistic situations, like labor mobility.

The pension systems built-in 1986 is targeting employees in enterprises, mainly in the urban area. The rural residents and urban residents without formal employment have no access to any pension in the form of social insurance. To improve their pension benefits, China introduced the basic old-age insurance systems for them in 2011. Individuals voluntarily join the system and pay a fixed amount annually into their individual accounts with government subsidies. Although it covers more people recently, the replacement of this pension system is very low.

In conclusion, China's current pension systems are targeting all types of residents. It includes three different schemes: Urban Employees' Scheme (UES), Basic Residents' Scheme (BRS), and Public Employees' Scheme (PES). The contribution rates are the same for UES and PES, 20% for employers, and 8% for employees. However, the benefit level varies significantly (Yuan, 2020). The UES is targeting a replacement ratio of 52.9% for an employee with 35 years of contribution history. While the BRS only has a replacement ratio between 15% and 30%. The PES is most generous with a replacement ratio ranging between 80% to 90%.

The pension system initially only targets the employees in the urban area. However, it provides large support of the retired life for the urban employees. Facing the population aging problem, China is actively reforming the social security system. The coverage and benefits level of the pension system is improving year by year. Therefore, it is important

to account for the pension system to analyze household savings in China, especially for the future.

2.3 Savings and Human Capital

Together with the rapid changing of the demographical structure in China, its saving rate changes as well. China's saving rate is relatively stable in the period of the 1960s-1970s. The gross saving rate is fluctuating around 35% between 1965 to 1980. The saving rates have gradually increased since the early 1980s after the One-child policy is implemented across the country. Figure 2 (a) shows the gross domestic savings over the period 1980 to 2015. It reaches more than a half in 2010, the highest point. Until 2015, it is still above 45%, among the highest in the world.

The framework I use includes several mechanisms through which fertility change to affect individual and aggregate savings and human capital.² First, there is a composition effect, through which the demographic structure will affect aggregate savings directly. As aggregate savings are mainly determined by the unconsumed labor income. A working-age population share change will lead to the aggregate savings change in the same direction. Second, the transfer effect of children's support for elderly parents. A reduction in fertility will reduce old-age support from children. This will increase private savings to finance old-age consumption. Third, the expenditure effect which means fewer children will take less resource from working households and leave more income to save.

Since 1960, another important fact is that China's human capital increases rapidly during the same period. Figure 2(b) shows the trend of average years of schooling for the population age 15 to 24. It gradually increases from 5 years to 9.5 years. Generally, education level increases together with the economy develop. However, in China transfer from children is an important source of old support. Human capital investment offers an alternative of saving to secure the old life when the number of children reduces. Fertility control from the one-child policy may give the incentive to invest in children instead of saving.

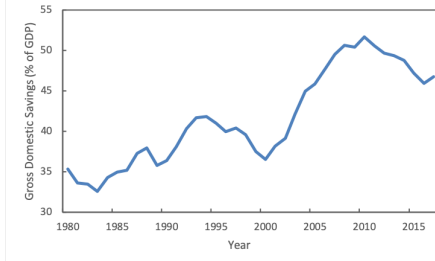
The pension system is not well developed in China. However, since the later 1980s, the enterprise employees have access to pension benefits after retirement. It accounts for part of their living expenditure. Thus, it will affect the savings for the old directly. Besides, as pension replaces part of the role of children in supporting the old-age, the developed pension system with wider coverage and better benefits may affect fertility and human capital investment decisions.

In the next section, I will discuss the model based on Choukhmane et al. (2013) with some modification, which captures all those mechanisms linking fertility with demographics, pension system, savings, and human capital accumulation decisions.

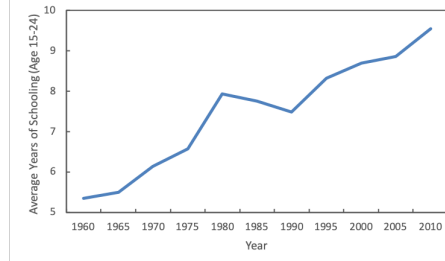
²I will briefly summarize the channels discussed in the literature here. The detailed theoretical analysis can refer Banerjee et al. (2014) and Choukhmane et al. (2013).

Figure 2: Savings and Human Capital in China Over Time

(a) Savings



(b) Human Capital



Data Source: Group et al. (2017); Barro and Lee (2013)

3 Model

In this section, I construct a multi-period overlapping generations model to evaluate the effects of the One-child policy and pension system on China's saving rate. The model is an extension of the one in Choukhmane et al. (2013). By adding the pension system, I can examine the interactions between fertility, human capital, and the pension system. China's fertility rate decreases after the One-child policy. Together with increasing life expectancy, China is facing a fast population aging problem. Although family support plays an important role in the old life, the traditional norm is facing challenges from declining fertility rates. The Chinese government has been taking reforms of the pension system and relaxing the One-child policy. The effect of those policies can be examined using this model, which incorporates the pension system, endogenous fertility, and human capital investment. It can help us understand the mechanisms of their effects on the economy and the interaction among them.

3.1 Households

The economy is populated by overlapping generations of individuals. Each individual lives up to J years and will die for sure in the last year. During the first J_c years, individuals are economically inactive. They will fully depend on their parents' support, making no choices and deriving no utility. After J_c years, they become an independent adult and start working. Agents' preferences are defined over consumption and follow an additive time separate utility function given by:

$$\sum_{j=1}^J \beta^{j-1} u(c_j) + v \log(n) \quad (1)$$

where β is the discount factor, c is consumption, and n is the number of children they have in their lifetime. $v > 0$ represents the preference over children. j denotes the number of years since agents become independent.

The fertility is determined by

$$u(c) = \log(c)$$

At age J_n ($J_n > J_c$), agents start supporting children. And to simplify, I assume they will bear all the children at the same time. Agents are actively working till age J_r ($J_n < J_r < J$). I abstract from modeling endogenous retirement. Since China has a mandatory retirement policy, the assumption of exogenous retirement at some age seems reasonable. The agents will die for sure at age J .

During working time, agents will provide inelastically one unit of labor supply and earn wages. The wages are subject to proportional social security taxes. Once they have children, they derive utility from them and support their consumption until they become independent. After retirement, agents will receive pension benefits from the government until they die. In the meantime, their children will give them money transfers to support their living as well. Children can also provide insurance for old-age consumption. So the budget constraints agents face every period is:

$$\begin{aligned} c_{j,t+j} + a_{j+1,t+j+1} = & a_{j,t+j}(1 + r_{t+j}) + 1_{\{j < J_r\}} \cdot w_{j,t+j}(1 - \tau_{t+j}) \\ & + 1_{\{j \geq J_r\}} \cdot b_{j,t+j} - T_{j,t+j} \end{aligned} \quad (2)$$

where $t + j$ represents time, t is the period when agents become independent and start working, and j represents agents age. Note that here I start counting ages from when the agents become independent. So the real age for an agent noted as j should be $j + J_c$. I will follow this notation rule thereafter. Therefore, $c_{j,t+j}$, $a_{j,t+j}$, and $w_{j,t+j}$ represents consumption, asset holding, and wage rate at time $t + j$ for agents born at $t - J_c$ respectively. $T_{j,t+j}$ denotes the intergenerational transfer between agents, and their children or parents. τ_{t+j} is the social security tax rate at period $t + j$. $b_{j,t+j}$ represents the pension benefits agents received from the government after retirement.

Wage earnings for each agent are determined by their age-specific productivity and human capital:

$$w_{j,t+j} = w_{t+j} e_j h_t^\alpha \quad (3)$$

where w_{t+j} represents the wage rate per efficiency unit at time $t + j$. I assume it follows a constant growth rate g_z . e_j denotes the efficiency units of labor for a worker with j years of working experience. And it is zero if retired. h_t represents the human capital specific to the cohort becoming an independent adult in period t . I assume the diminishing marginal return of human capital.

The human capital of each agent is determined by his parent's human capital and their investment. To simplify, I assume the human capital is fixed after the agents become independent. The human capital of agents starts working at period $t + J_n$ is:

$$h_{t+J_n} = h_t^\rho (i_t^h)^{1-\rho} \quad (4)$$

where $\rho \in [0, 1]$ measures the intergenerational transmission of human capital. $\rho = 0$ means there is no inheritance of human capital.

The intergenerational family supports are denoted by $T_{j,t+j}$. It includes the costs for agents taking care of their children before they become independent, the support of their parents after they retired, and the money transfer they receive from their children once retired. The expenditure on children not only involves the consumption need of them but also the education cost. Following the paper by Choukhmane et al. (2013), the time and scale of expenditure on children is revealed from microdata. The expenditure of rearing and educating children is a fraction of the agent's wage income. I assume the costs are paid from age J_n to J_e . It is a function of the human capital investment on children $\phi_j + \phi_j^h i^h$. The first part represents the fixed cost of having children, like mouth-to-feed expenditure, compulsory education, and so on. The second part denotes the discretionary education cost, which is determined by the agent's human capital investment decision on children. Another important fraction of the transfer is the support from agents to their retired parents. Similar to before, I assume it is a fraction of their wage rates. The last fragment of the transfer is the support agents received after retirement. To be consistent with the assumption above, it is a fraction of their children's wage rates. Therefore the total transfer from agents aged j at period $t + j$ is determined as

$$T_{j,t+j} = \left[1_{\{J_n \leq j \leq J_e\}} (\phi_j + \phi_j^h i_t^h) n_t + 1_{\{J_r - J_n \leq j \leq J - J_n\}} \psi \frac{n_{t-J_n}^{\omega-1}}{\omega} \right] w_{j,t+j} \quad (5)$$

$$- 1_{\{J_r \leq j \leq J\}} \psi \frac{n_t^\omega}{\omega} w_{j-J_n, t+j}$$

where ϕ_j and ϕ_j^h represents the costs of raising children and educating children in terms of agents' wage rates respectively and ψ represents the transfer in terms of wage rates to agents' retired parents. And I assume there exists a possibility of free-riding among siblings to share the burden of transfers following the Choukhmane et al. (2013). That means the transfer is decreasing in the number of siblings. It is captured by ω ($\omega < 1$). I abstract from modeling the altruism on retired parent's utility. It is captured by the parameter ψ . This makes it easier to model the fertility choice with the intergenerational transfer. As in this setup, children can be an investment of old-age insurance. Note that I assume the total cost of having children is linear in the number of children. I abstract from cost sharing from raising multiple children. One reason is that the cost of raising children revealed from data is mostly education cost, which is not shareable. The mouth-to-feed cost just accounts for a small fraction. Therefore, I assume there is no cost sharing among siblings.

3.2 Government

I assume a Pay-as-you-go (PAYGO) pension system in the economy. The government is collecting a social security tax to finance the pension system. The contribution tax rate is fixed at τ percent. And I assume a balanced budget for the government every

year, which means the pension benefits equal to the total contributions every year. The benefit rate is endogenously determined by the tax revenue, which, in turn, is affected by the demographic structure. Given the demographic transition and high wage growth in China, in the beginning, this pension system will give high pension benefits for the earlier generations. Once the fertility decreases and the population starts aging, the replacement rate is smoothly decreasing. The pension benefit b_t at period t is determined by

$$b_t = \frac{\tau \sum_{j=0}^{J_r-1} \mu_{j,t} w_{j,t}}{\sum_{j=0}^{J_r-1} \mu_{j,t}} \quad (6)$$

where $\mu_{j,t}$ represents the population share of age j at period t . $w_{j,t}$ represents the wage rate for agents aged j at period t , which is determined by Equation (3).

3.3 Household Decision Problem

At every period, the agents need to decide the amount of consumption and the next period's assets given wage rate, interest rate, present assets holding, and net income after tax and transfer. When agents have children, they also need to make decisions of human capital investment on their children other than feeding them. So the decision problem agents facing every period except period J_n is

$$V_j(a, n, h) = \max_{a', c} u(c) + \beta V_{j+1}(a', n, h) \quad (7)$$

subject to

$$c + a' = a(1 + r) + 1_{\{j < J_r\}} \cdot w(1 - \tau) + 1_{\{j \geq J_r\}} \cdot b - T \quad (8)$$

where w represents the wage rate of agents aged j with human capital h . It is determined by Equation (3). And T represents the net transfer from agents aged j with human capital h . It is defined as Equation (5). b denotes the pension benefits agents receive at age j . It is determined as Equation (6).

At period J_n , the agents will have children. They need to make decisions about the number of children and the amount of human capital investment on them, as well as consumption and assets holding. So the decision problem of the household at period J_n is:

$$V_j(a, n, h) = \max_{a', c, n', h'} u(c) + \beta V_{j+1}(a', n, h)$$

subject to the same budget constraint as above, Equation (8).

By introducing the endogenous human capital decision from parents on children, I can capture the relationship between the pension system and fertility and human capital. Given the important role of family support for the old-age in China, children will provide insurance for old-age consumption. Investing in children's human capital is one way to ensure retired life. The one-child policy restricts the insurance provided by children from reducing the number of children. Households can invest in children's human capital to

compensate for the effect of this change. As a result, the human capital of only children may increase due to the One-child policy. Modeling the fertility and human capital decision jointly can capture this quantity-quality trade-off considering old-age support.

In the meantime, China has been reforming its social security system during the past decades. The pension system provides insurance on the retired life. A well-functioning pension system will reduce the old-age provision saving incentive. It replaces the role of children regarding supporting the old. This will affect the investment decision from parents on children as well. One thing to note here is that I abstract from the altruism of parents on children’s utility. The motivation to invest in children’s human capital is just to secure old-age consumption. Therefore, the effect of the pension system on human capital investment only reflects the channel through old-age support replacement.

3.4 Mechanisms of Effects on Saving

There are four channels included in this model through which demographics can affect household saving as discussed in Choukhmane et al. (2013).³ The first channel is the “expenditure channel”. One-child policy limits the number of children in each household to be no more than one. That means less expenditure is needed on dependents, thus more resources left for the middle-aged parents to save. The second channel is the “transfer channel”. In China, family support is the main source of retirement income. Declining fertility rate means less support for the retired. Therefore, the middle-aged would like to save more for the old. The third channel is the “human capital investment channel”. Instead of saving, the middle-aged can also invest in children’s human capital to secure the old age. Higher human capital gives children a better ability to support families, which can compensate for the loss from a smaller number of children.

The fourth channel is the “composition channel”. On one hand, the saving rate varies across the life cycle. Usually, the saving rate of the middle-aged is higher than the others. On the aggregate level, demographic composition change will change the fraction of the saving population, thus affect the aggregate saving rate. On the other hand, demographic composition change means the fraction of working-age population changes, which will affect the capital-labor ratio. Accordingly, the wage rate and interest rate will be affected. Changes in labor income and saving return, in turn, will affect household saving decisions.

Considering the pension system, it can directly affect households’ saving decisions. In this paper, I abstract from labor income uncertainties. The saving incentive is mainly to smooth consumption over time, especially for the retired period. Pensions provide insurance for old-age life. Sufficient pension benefits will reduce households saving incentive and stimulate consumption.

In the meantime, pension benefits will interact with demographics and therefore affect saving decisions of households. For instance, there are literature shows that the pension system will affect fertility decision due to “old-age security” (Billari and Galasso, 2009; Shen et al., 2020). The children can provide support for old-age life. One reason for

³I will highlight the primary mechanisms. For more details about the theoretical analysis, please refer Choukhmane et al. (2013).

households to have children is to provide security of consumption in retired life. Similarly, pensions offer insurance for old-age consumption as well, which can supplement the support of children. Pension benefit changes will affect the insurance of the old. Therefore, it will affect the “old-age security” incentive for households to have children. Besides, human capital investment is another way for parents to invest in children and ensure old-age life. The one-child policy limited the number of children each household can have. Alternatively, households can invest in children’s human capital to make up for the loss from the number of children reduction. Following the same logic, pension benefit changes will also affect households’ decisions on investment in the human capital of children.

On the other hand, the pension system will be affected by the demographics changes. For example, the pension benefits in a pay-as-you-go pension system are determined by the contribution population and retired population. With an aging population, the fraction of the retired will increase compared to the fraction of pension contributors. This will reduce the pension benefits, which, in turn, will affect savings, fertility decisions, and demographic structure.

By building this model with the pension system, I can examine the contributions of demographics and pension on saving rates respectively. Also, it can show the interactions between pension and demographics as well.

4 Parameterizing the Model

To examine the linkages between fertility, pension system, savings, and human capital, I use a parameterized version of the model. In the exercise, I take some of the parameter values directly from the literature or real data about China. The more accurate and careful calibration will be future work. I will discuss the details about the parameterization in the following.

4.1 Demographics

Each model period is corresponding to 4 calendar years. I model the whole lifecycle period of an individual from ages 0 to 80, to capture the demographics accurately. The maximum agent age index is $J = 20$, which means the agents will die for sure at age 80. I abstract from modeling mortality for different ages. Although this may lead to deviation from real life expectancy over time, the main purpose of this paper is to examine the effects of fertility on savings, human capital, and its relationship with the pension system. This simplification can make us focus on the mechanisms through which fertility affects the economy. During the first $J_c = 5$ periods, agents are dependents of their parents and make no decisions. This represents that the agents become economically independent and working alone at age 17-20. I set $J_n = 7$, which implies agents will have children at real age 25-28. Agents will retire and start receiving pension benefits and children’s transfer at period $J_r = 15$, corresponding to the forced male retirement age, 60, in China.

The initial distribution of the population in 1964 is set to match the size of each age

group above 17 years old in 1982.⁴ This is to make sure the composition effect on aggregate savings are in line with the population distribution when the one-child policy is implemented. The following population of each age group evolves consistently with the path of fertility in the model. And it is determined as:

$$N_{j-J_n,t} = N_{j,t} * n_{J_n,t-j+J_n}$$

where $N_{j,t}$ represents the number of agents aged j at period t , $n_{J_n,t}$ represents the fertility of agents aged J_n at period t . One thing to note here is that the fertility rate will change according to time. In the initial steady state, the fertility equals to 3, which is corresponding to the average fertility over period 1964-1969. And the preference parameter for children v is calibrated to match the steady state (Choukhmane et al., 2013).

The One-child policy is introduced across the whole country in 1979. To model the policy changes, the fertility choice will be constrained by one once implemented. Besides, there already exists a family plan policy before the universal One-child policy to restrict the number of children each household can have. To capture the effect of those policies, the specific restrictions on the number of children will be put for each year after 1970.

4.2 Pension System

The current Chinese pension system is first introduced in 1986 and under major reform in 1997. The contribution rates are 20% for the employers and 8% for the employees. In 2011, it develops the basic pension system to cover all types of the population. Although there are some variations in benefits and replacement rates among those pension systems, the contribution rates are generally the same for households. To simplify, I assume a Pay-as-you-go pension system in the economy, which is funded by the labor income tax. The tax rate is set as $\tau = 20$ to match the contribution rates. And it is introduced after the One-child policy. The coverage of the pension system will change according to the policies.

I abstract from modeling rural and urban differences. The main focus here is to show the effect of the pension system on savings, and its iteration with fertility and human capital. To capture the effects of the pension system on aggregate savings accurately, a careful model of the pension system and rural-urban migration overtime is needed. I leave this for future work.

4.3 Parameters

Most parameters are borrowed from Choukhmane et al. (2013). I will briefly introduce the parameters and the way to calibrate their values in this section. More details will be discussed in the paper.

The wage rates are considered growing with a constant rate $g_z = 6.1\%$ to capture the high growth rate of the Chinese economy from the 1980s to 2010s. The production

⁴This setup is for modeling the transition dynamics since 1964. This paper does not accomplish this step. It will be future work.

Table 1: Model Parameters

Parameter		Value
r	Real interest rate	5.3%
g_z	Wage growth rate	6.1%
α	Technological parameter	0.37
v	Children preference	0.58
ω	Free-ride parameter	0.65
ψ	Old parents altruism parameter	9%
β	Discount factor (annual)	0.99
ρ	Human capital inheritance parameter	0.2

technology parameter α is set to 0.37. Experience parameters are shown in Table 1, which is estimated in their paper using micro survey data about China. The real interest rate of saving faced by households is relatively stable over time in China, which is set to be 5.3%.

The parameters related to old age support are ψ and ω . The first one captures the degree of altruism toward the retired parents. The second one reveals the propensity of free-ride for an agent on transfers to parents provided by siblings. Following Choukhmane et al. (2013), I set ω equals to 0.65, which is estimated empirically using micro survey data. While ψ will be calibrated using the model to match initial steady state variables in the economy.

Parameters related to the costs of raising and educating children are shown in Table 2. (Choukhmane et al., 2013) estimates those using the education expenditure survey data about China. Once the other parameters are fixed, they calibrated the rest of the parameters $\{\beta, \psi, \rho\}$ to match the model predicted age-saving profile and education expenditure with the microdata. The values of all the parameters except the ones related to raising costs and experience are shown in Table 1.

5 Results

In this section, I discuss the implications from the model about the effect of fertility and pensions on savings, human capital and their interactions between each other.

5.1 Micro-level Predictions

In this section, I will show the theoretical model implications of the effect of fertility and pension benefits. To be more specific, I compare the model results for three different scenarios: the one without fertility restrictions (or the One-child policy (OCP)), the

Table 2: Parameters of Costs and Experience

Period	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
e_j	0.64	0.70	0.86	0.89	0.91	0.98	1.00	0.88	0.73	0.51	0.15	0.10	0.17	0.03	0.02
ϕ_j		0.02	0.08	0.09	0.12	0.12	0.09	0.03	0.02						
ϕ_j^h			0.00	0.01	0.01	0.04	0.07	0.04	0.02						

Table 3: Age-saving Profiles Comparison

Saving rate		Non-OCP	OCP	Difference	OCP&Pension	Difference
$j = 7 - 8$	(25-32)	6.4%	12.6%	6.1%	2.4%	-10.2%
$j = 9 - 10$	(33-40)	4.3%	18.9%	14.6%	0%	-18.9%
$j = 11 - 12$	(41-48)	22.5%	37.6%	15.0%	1.6%	-35.9%
$j = 13 - 14$	(49-56)	44.1%	44.2%	0.1%	9.9%	-34.2%

one with fertility restrictions, and the one with fertility restrictions plus pension system. I need the demographic structure to get pension benefits. To simplify, I assume the population of each age group is evenly distributed. This means the replacement ratio will be about 40% according to Equation 6. This assumption will not affect the theoretical implications. However, I need to calibrate the demographic structure to examine the effects of the One-child policy and pension system on savings and human capital in China. This will be in the next step.

5.1.1 Micro-level Effects

This section is to examine the effect of the One-child policy and pension benefits on individual household's saving and human capital decisions. I assume all the households have 0 initial assets and compare the choices with and without those policies.

Table 3 shows saving rates for households of different age groups for the three scenarios. The One-child policy increases savings for all the age groups and it leads to a relatively flat age-saving profile for the middle-aged (the 20s to 50s). This is consistent with the predictions in Choukhmane et al. (2013). In their experiment with twins, the model predicts 8 percentage points increase in saving for the agents aged between 41 to 48, which is the highest among all the age groups. This is corresponding to 7-10 percentage points increase in for the same age group in empirical data. My predictions are higher than their work. Considering the different assumptions of initial conditions, the predictions are reasonable.

Comparing scenarios with and without the pension system, the model predicts that pension benefits will reduce the saving rates across all age groups. This is consistent with empirical work about pension and savings. For instance, Feng et al. (2011) shows that pension wealth will offset household savings significantly.

Table 4 shows the education expenditures responses of households to different policies. The One-child policy will reduce the total education expenditures of parents on children. This reflects the expenditure channel through which fertility will affect household savings. With fewer children to support under the One-child policy, households have more resources left to save.

Comparing the human capital investment on each child, the One-child policy leads to human capital increases by 42% for the only child. This is consistent with the result

Table 4: Education Expenditure Comparison

Education Expenditure (% of Wage Income)		Non-OCP	OCP	Difference	OCP&Pension	Difference
$j = 7 - 8$	(25-32)	8.9%	4.0%	-4.9%	4.0%	0%
$j = 9 - 10$	(33-40)	14.9%	7.0%	-7.9%	6.7%	-0.3%
$j = 11 - 12$	(41-48)	20.8%	11.4%	-9.4%	9.3%	-2.0%

from Choukhmane et al. (2013). Their work predicts that the human capital of the only child will increase by 53% due to the policy. This reflects the quantity-quality tradeoff. Investing in children is an alternative of savings in terms of old-age support. When the number of children decreases, the expectation of future income for the old will decrease as children are an important source for old-age support. Households can invest more in children's human capital so that each child can offer better support when they retire. Besides, total education expenditure reduces even though the human capital investment on one child increases after the One-child policy. It shows that investment in children's human capital cannot fully offset the effects of the reduction in the number of children.

And for the pension system, the model predicts it will reduce the total education expenditure. Given that the fertility choices are binding to the constraint, this difference is coming from less investment in children's human capital. Comparing human capital investment choices, this PAYGO pension system reduces children's human capital by 29%. It shows that improvement in public support of the old may cause less human capital accumulation in the society with filial piety.

The model predictions are reasonable comparing with relative research. However, it still needs careful calibration to investigate the effects of One-child policy and pension system and empirical tests to verify the predictions.

5.1.2 Households with Different Initial Wealth

The model can generate rich micro-level predictions. In this section, I will show variances in response to the One-child policy and pension system for households with different initial wealth using the model simulation results.

Table 5 the age-saving profiles for households with different initial assets. First, the One-child policy increases the saving rates for households with different initial wealth in general. The effects variations for different age groups. Saving rates for young parents (less than 40) increases less than older parents (less than 50). One reason is that young parents need to support children and have fewer resources to save. However, for agents close to retirement the increase is small. The model implication is consistent with the results in Choukhmane et al. (2013). In their model simulation of the age-saving profiles in 1986 and 2009, the saving rates are generally increasing over the life cycle till age 50. And there is an upward shift for the age-saving profiles between 1986 and 2009. The

Table 5: Age-saving Profiles for Households with Different Initial Assets

		$a_0 = 0.4$			$a_0 = 1.5$		
Saving rate		Non-OCP	OCP	OCP&Pension	Non-OCP	OCP	OCP&Pension
$j = 7 - 8$	(25-32)	6.3%	13.7%	0.8%	10.7%	13.6%	-10.2%
$j = 9 - 10$	(33-40)	3.5%	19.1%	0%	4%	23.9%	-16.0%
$j = 11 - 12$	(41-48)	21.6%	40.3%	1.6%	9.2%	36.6%	-3.1%
$j = 13 - 14$	(49-56)	44.1%	44.2%	9.9%	44.2%	41.5%	5.8%

middle-aged (30-40) saving rate increases more than the other age groups.

Second, the pension system will discourage private savings from households. Pension benefits provide insurance for old-age consumption. When pension benefits increase, there is less necessity of saving to provide old-age life support. This prediction is consistent with the empirical work which shows that pensions will substitute the other forms of private saving (Bosworth and Burtless, 2004; Attanasio and Rohwedder, 2003; Attanasio and Brugiavini, 2003). Besides, the model predicts wealthy households will save much less with pension benefits increase. In this framework, I assume no bequests, which means households will hold zero assets in the end. With pension benefits insuring the old-age life, the wealthy households need to reduce more asset holding to smooth their life-cycle consumption. One thing to note, all the model implications are based on the assumption of an evenly distributed population and PAYGO pension system. To examine the effect of China's pension system, I need to model the demographic structure in China and model the pension system more carefully.

Besides, the saving rates for the wealthy households predicted in the model contradict the data. For instance, in the U.S. data, rich people keep saving at high rates. Without modeling the bequests motive, it is hard for the life-cycle to match the high saving rates for the rich (De Nardi and Fella, 2017). To examine the household saving rates in past China, this may not be an issue. As most Chinese families are very poor and not much wealth left to the next generation. However, it may matter for recent years with high economic growth and increasing wealth inequality in China. According to Choukhmane et al. (2013), their model fails to match the high old-age saving rates in 2019 for Chinese households. The main purpose of this paper is to examine the aggregate household saving rates. I will leave the old-age and rich household savings to future work.

For human capital investments, the model results show that households will invest more in children after the One-child policy. But it does not predict many variations among households with different wealth. Pension benefits reduce the household investment in children's human capital, but mostly for the poor ones. The prediction of the model is consistent with studies about the pension system and human capital investment. For instance, Sinn (2004) shows that the PAYGO system can serve as insurance of having no children and it reduces the incentive to invest in human capital.

The model also predicts variation in fertility choices among different households. The

wealthy families are willing to have more children if there is no fertility constraint according to the model. These results are in line with empirical studies that show a positive relationship between wealth and fertility (Lovenheim and Mumford, 2013; Hopcroft, 2019).

In conclusion, the model predicts that the One-child policy will increase saving rates for all age groups and pension benefits will have the opposite effects. But the responses to pension benefits vary across households with different wealth. Human capital investment will increase with the One-child policy. But pension benefits increase can discourage parents from investing in children. One thing to note here is that all those results are predicted based on the model. To examine the real effects, I need to combine it with the real data.

5.2 Aggregate Savings and Human Capital

This section is to examine the aggregate effects of the One-child policy and pension system. The population distribution is following the assumption of the last section. This is just to display the potential effects of those policies. To quantitatively investigate their effects, I need to calibrate the population distribution before the One-child policy is implemented and simulate the changes accordingly.

The aggregate saving rate is defined as

$$S_{j,t} = \frac{\sum_{j=j_c+1}^J \mu_{j,t} s_{j,t}}{\sum_{j=j_c+1}^J \mu_{j,t} y_{j,t}}$$

where $s_{j,t}$ represents the savings of households aged j at period t , which is determined by the asset holding changes across two periods. $\mu_{j,t}$ is the population share of age group j at period t , and $y_{j,t}$ is the after-tax income.

The aggregate saving rates are displayed in Table 6. On the aggregate level, the One-child policy increases the saving rates by 6.2 percentage points. This is based on the assumption of an evenly distributed population. However, the working-age population accounts for more than half of the total population in China since the 1970s. Therefore the effects are expected to be larger.

Adding the pension system makes the aggregate saving rate decreased to negative. Under the assumption of population distribution and pension system, the pension benefits are high. And it will reduce the saving rate for all age groups and lead to negative saving rates for the retired. The large fraction of the retired population may lead to the aggregate saving rate to be negative. With population aging and pension reforms undertaking in China recently, the future aggregate saving rate is expecting to decrease following the predictions of the model.

Besides, the model predicts the human capital of the only-child generation increases by 71% compared with their parental generation. This corresponds to a wage increase of 22%. While under the assumed pension system, the human capital of the only-child generation will only increase by 30%. This shows that pension benefits lead to less human capital accumulation on the aggregate level. One thing to note, I assume no altruism

Table 6: Aggregate Savings

	Non-OCP	OCP	Difference	OCP&Pension	Difference
Aggregate Saving Rate	9.2%	15.4%	6.2%	-19.5%	-34.9%

from parents on children’s utility in this setup. The only incentive to invest in children’s human capital is to secure old-age consumption. Therefore the model can only reflect the effects on human capital accumulation through this mechanism. Besides, this prediction is based on the assumption of fixed population distribution. As we all know, fertility and population structure determine pension benefits. Changes in pension benefits, in turn, will affect fertility and human capital decisions. With this population assumption, the prediction of pension effects abstracts from the composition effects. China has introduced a pension system targeting all the population in 2011. In the meantime, it has relaxed the One-child policy in 2015. To examine the effects of those reforms, I need to model the endogenous population distribution. And this will be the next step.

6 Summary and Future Work

The gross saving rate has increased drastically in China since the 1970s. After introducing the One-child policy, the fertility rate declines radically. As children are an important source for old-age support, the decrease in the number of children will lead to a reduction in expectation of future transfer for the old-age. Therefore, the One-child policy will increase the incentive of saving for the old-age. Pension benefits offer insurance for old-age consumption. Therefore, it may discourage households’ saving for old-age. This paper investigates the effect of the One-child policy and pension system reforms on China’s saving rates using an overlapping generations model. The model incorporates intergenerational transfers, endogenous fertility, human capital investments, and the pension system.

The results show that the One-child policy will increase individual household saving rates and investment in children’s human capital. While the pension system will discourage households’ saving and investment in children. However, the response varies across households with different wealth. Also, the model predicts a positive relationship between wealth and fertility, which may indicate heterogeneity in welfare effects. And the model fails to explain the wealthy households’ high saving rates. On the aggregate level, the model shows that the One-child policy will lead to an increase in saving rates and higher human capital accumulation of the only-child generation. Pension systems on the other hand will discourage savings and human capital investment. That indicates negative long-run effects on the economy.

Given those implications from the model, there are two types of future work to be done. The first type is to complete the analysis of the effects based on the current framework. To be more specific, I need to calibrate the model to the Chinese economy carefully and examine the micro-level and aggregate effects of the One-child policy and pension systems. This can also generate the implications of relaxation of the One-child policy

and pension reforms in China. And to examine the effects of aggregate savings, I need to simulate the transition dynamics of saving rates and decompose the aggregate saving rates to compare the effects of different forces. Besides, I need to test the model implications with empirical work using microdata. The second type of work is to solve the limitation of this work. One thing is that the framework is modeling the intergenerational transfer exogenously given. That makes it easy to model fertility decisions. However, family support will respond to pension benefit changes in reality. It may have implications on the economy with pension reforms. The second thing is that I abstract from rural-urban migration. The pension reforms have distinct effects on rural-urban households. And the rural-urban population structure has changed a lot in China. There will be different effects of pension reform between rural and urban households. The third thing is that the model does not include any risks. Rural and urban households are facing different health and income risks. Their response to fertility restrictions and social security reforms may be different.

References

- Attanasio, O. P. and Brugiavini, A. (2003). Social security and households' saving. *the Quarterly Journal of economics*, 118(3):1075–1119.
- Attanasio, O. P. and Rohwedder, S. (2003). Pension wealth and household saving: Evidence from pension reforms in the united kingdom. *American Economic Review*, 93(5):1499–1521.
- Bairoliya, N., Canning, D., Miller, R., and Saxena, A. (2018). The macroeconomic and welfare implications of rural health insurance and pension reforms in china. *The Journal of the Economics of Ageing*, 11:71–92.
- Bairoliya, N., Miller, R., and Saxena, A. (2017). The macroeconomic impact of fertility changes in an aging population. *Available at SSRN 3016158*.
- Banerjee, A., Meng, X., Porzio, T., and Qian, N. (2014). Aggregate fertility and household savings: A general equilibrium analysis using micro data. Technical report, National Bureau of Economic Research.
- Barro, R. J. and Lee, J. W. (2013). A new data set of educational attainment in the world, 1950–2010. *Journal of development economics*, 104:184–198.
- Billari, F. C. and Galasso, V. (2009). What explains fertility? evidence from italian pension reforms.
- Bosworth, B. and Burtless, G. (2004). Pension reform and saving. *National Tax Journal*, pages 703–727.
- Choukhmane, T., Coeurdacier, N., and Jin, K. (2013). The one-child policy and household savings.
- Curtis, C. C., Lugauer, S., and Mark, N. C. (2015). Demographic patterns and household saving in china. *American Economic Journal: Macroeconomics*, 7(2):58–94.
- De Nardi, M. and Fella, G. (2017). Saving and wealth inequality. *Review of Economic Dynamics*, 26:280–300.
- Feng, J., He, L., and Sato, H. (2011). Public pension and household saving: Evidence from urban china. *Journal of Comparative Economics*, 39(4):470–485.
- Group, W. B. et al. (2017). *World development indicators 2017*. World Bank.
- Hopcroft, R. L. (2019). Sex differences in the association of family and personal income and wealth with fertility in the united states. *Human Nature*, 30(4):477–495.
- İmrohoroglu, A., Zhao, K., et al. (2017). Household saving, financial constraints, and the current account in china. Technical report, Working paper.

- Lovenheim, M. F. and Mumford, K. J. (2013). Do family wealth shocks affect fertility choices? evidence from the housing market. *Review of Economics and Statistics*, 95(2):464–475.
- Shen, Z., Zheng, X., and Yang, H. (2020). The fertility effects of public pension: Evidence from the new rural pension scheme in china. *Plos one*, 15(6):e0234657.
- Sin, Y. and Yu, X. (2005). China-pension liabilities and reform options for old age insurance. Technical report, The World Bank.
- Sinn, H.-W. (2004). The pay-as-you-go pension system as fertility insurance and an enforcement device. *Journal of Public Economics*, 88(7-8):1335–1357.
- Song, Z., Storesletten, K., Wang, Y., and Zilibotti, F. (2015). Sharing high growth across generations: pensions and demographic transition in china. *American Economic Journal: Macroeconomics*, 7(2):1–39.
- UN (2019). World population prospects 2019.
- Yuan, R. (2020). *Financial sustainability of the pension system in China: Impact of fragmented administration and population ageing*. PhD thesis.
- Zeng, Y. and Hesketh, T. (2016). The effects of china’s universal two-child policy. *The Lancet*, 388(10054):1930–1938.
- Zhang, J. (2017). The evolution of china’s one-child policy and its effects on family outcomes. *Journal of Economic Perspectives*, 31(1):141–60.